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NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2025-2026

SC2006 – SOFTWARE ENGINEERING

Apr/May 2026 Time Allowed: 2 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 4 pages.
2. Answer **ALL** questions.
3. This is an open-book examination.
4. All questions carry equal marks.
5. Refer to Appendix A on page 4 for the project description which is needed to answer some of the questions.

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1. Based on the project description given in Appendix A,

   (a) Identify the actors and use cases of the project and draw the use case diagram. Use <<include>> and <<extend>> relationships where appropriate.

   (18 marks)

   (b) Draw an activity diagram with swim lanes if any, that shows the activities for the process of checking the JSON output.

   (7 marks)

2. Based on the project description given in Appendix A,

   (a) For the functionality of the recommendation generation, identify the main classes and their associations and draw a conceptual class diagram. Your conceptual class diagram should clearly depict the stereotypes of each class. You do not need to identify any attributes or operations within the classes.

   (8 marks)

Note: Question No. 2 continues on Page 2

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(b) Draw a state machine diagram for the freeform conversation session that shows the states and transitions of the conversation component.

(6 marks)

(c) One possible change in requirements during the project would be altering the contents of the **tag set**. Compare the Waterfall model and Agile methodology in their handling of this possible change in requirements.

(5 marks)

(d) A team chose the Scrum method to develop the given project. After 2 months, they found the sprints were frequently overrun. Analyze two possible causes and propose corrective actions.

(6 marks)

3. (a) Propose an appropriate architecture for the Restaurant Recommendation System as described in Appendix A.

(3 marks)

(b) Draw a detailed Class diagram with key attributes and methods for the Restaurant Recommendation System, reflecting your architecture choice in Q3(a).

(12 marks)

(c) When implementing the Restaurant Recommendation System, your company decides to pay for access to an externally hosted Large Language Model (LLM) in order to power the freeform conversation functionality.

(i) Your company could have instead chosen to host their own LLM. Describe TWO upsides and TWO downsides to this alternative approach.

(4 marks)

(ii) After deploying the system, you find that the externally hosted LLM is replying to users with advertisements for your competitors if it detects that it is being used by your service. Aside from your company hosting its own LLM, describe TWO other approaches your company could take to address this issue.

(6 marks)

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4. A helper function used in defining the system of Appendix A, `validateUserInput`, takes two arguments: a string representing the user's postcode, and an integer representing the user's maximum desired travel distance in kilometers.

The user's postcode must be a string of six numerical digits – e.g. "612345". If another string such as "a28c0cw" is provided, the system will reject the user's input.

Assume that the travel distance provided by the user is only permitted to be between 0 and 100 kilometers, and that other inputs such as -100 or 9999 will cause the system to reject the user's input.

(a) For the function described above, determine the equivalence classes for the above TWO inputs. State any assumptions you make.

(4 marks)

(b) Determine the boundaries of the equivalence classes identified in your answers to Q4(a), if they exist. For each boundary, identify a value on the boundary, a value just below the boundary, and a value just above the boundary.

(4 marks)

(c) You intend to perform **defensive testing** for the `validateUserInput` function described above. Design a set of test cases to test the TWO inputs based on the equivalence classes and boundary values identified in your answers to Q4(a) and Q4(b).

(8 marks)

(d) Imagine that you have been asked by your manager to modify the `validateUserInput` function, so it additionally validates that the input postcode corresponds to a real building in Singapore by querying a public online API provided by Singapore Post.

(i) Describe how this change would affect the equivalence classes of the function's postcode input.

(3 marks)

(ii) Discuss TWO ways in which this change would complicate the process of writing comprehensive tests for the function.

(4 marks)

(e) Imagine that you want to extend the system to be deployed in multiple countries, each with a different postcode format. Briefly discuss how the definition of `validateUserInput` might need to change to accomplish this.

(2 marks)

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Appendix A

Restaurant Recommendation System

You are building a system and web application that gives restaurant recommendations to users in Singapore. The system maintains a database of restaurants in Singapore. Each database entry includes the name of the restaurant, the address of the restaurant, the restaurant's website address, and a list of qualities such as "burgers", "spicy", "good for dates". These qualities must be selected from a fixed set of 20 tags that are determined during the system's initial design, known as the **tag set**.

When opening the web application, the user is asked to anonymously engage in a freeform conversation with a Large Language Model (LLM)-based chat interface about their food preferences. After the user has sent 10 messages and received 10 responses, or alternatively if they click a "finished" button before this, the conversation is terminated. With the context of the conversation, the same LLM is then asked by the system to produce a JSON output listing up to 5 tags from the **tag set** it believes the user is most interested in. The system checks that the output is valid JSON, contains only tags from the **tag set**, and contains no more than 5 tags. If any of these checks are not satisfied, an error message is displayed to the user and they are invited to begin the process again.

If all of the checks are satisfied, the user is asked to input their postcode, and an integer X. The web application takes the user's preference tags and displays a list of restaurants (with website links) from the system database that are no more than X kilometers from their provided postcode. The returned recommendations are sorted according to how many of the user's preference tags match the restaurant's tags (most to least). If the user clicks on the website link of a restaurant, the preferences of the user, along with their selected restaurant, are logged by the system.

An administrator login portal is provided by the system. An administrator can log in with a special username and password. The administrator can view the log of user-selected restaurants, and add, remove, or alter restaurant entries to/from/in the database.

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.